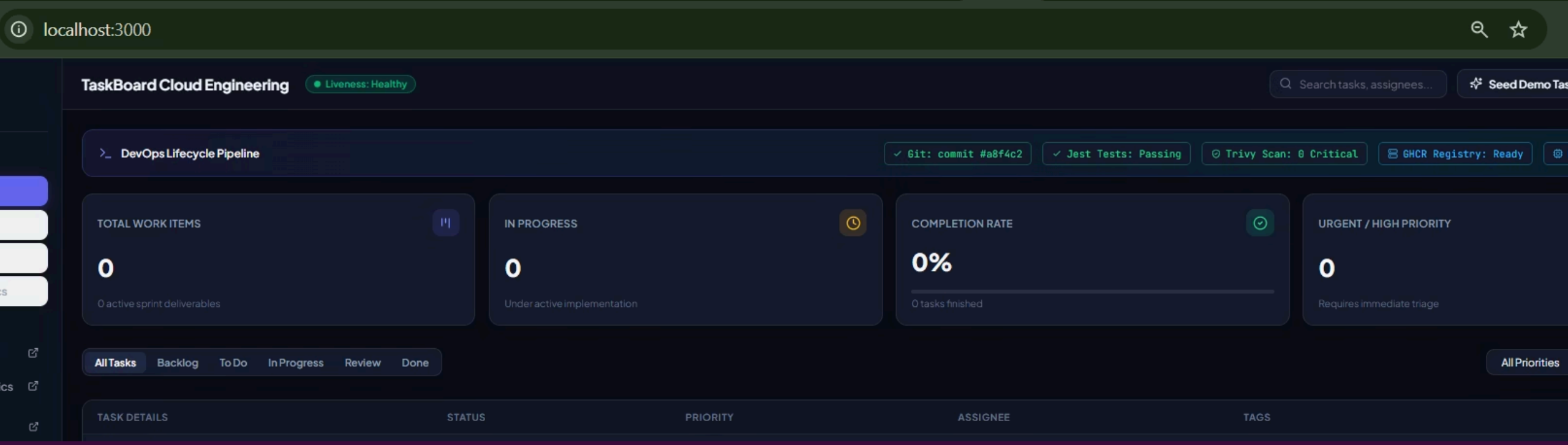


DEVOPS · CLOUD-NATIVE CAPSTONE

TaskFlow Cloud

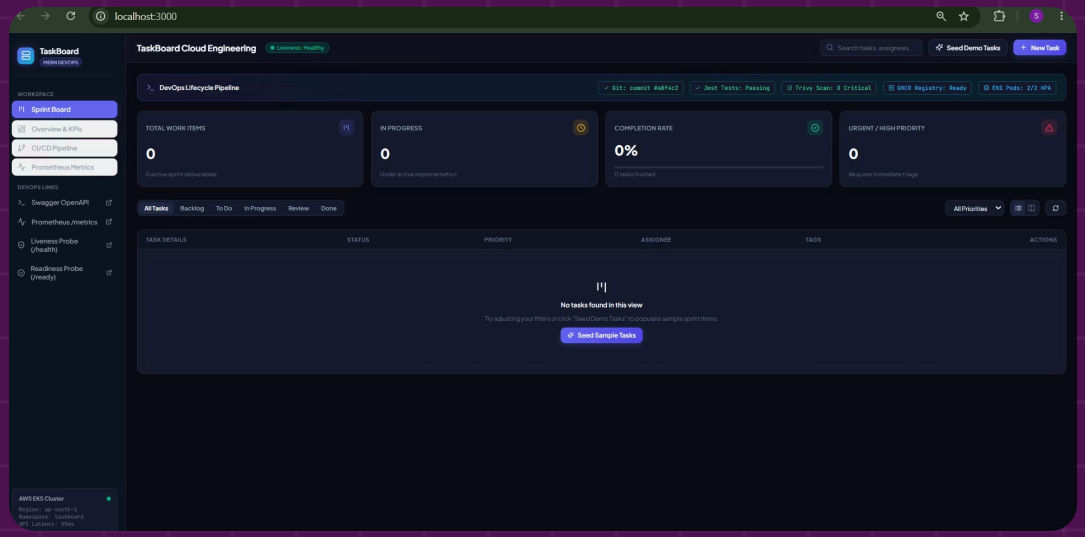
DevOps Cloud-Native Capstone

End-to-end MERN cloud platform architecture

Git → CI/CD → DevSecOps → Terraform → Kubernetes → Prometheus/Grafana

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PLACEHOLDER · TaskBoard UI screenshot



01 · APPLICATION

The application tier starts with operational readiness

Interactive Kanban board

Built with React, Vite, and a Node.js REST API.

Persistent sprint state

MongoDB persists sprint task state.

Native probes

/health and /ready support Kubernetes readiness.

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02 · QUALITY GATE

Automated tests create the first release gate

A green test run is the first permission to build.

01

Run

10 automated test cases across 2 suites.

02

Isolate

Mock database layer avoids live-data pollution.

03

Gate

100% green status; any failure halts downstream image builds.

PLACEHOLDER · Jest terminal screenshot

```
LENOVO@shreesha MINGW64 ~/TaskFlow-Cloud (main)
$ cd backend

LENOVO@shreesha MINGW64 ~/TaskFlow-Cloud/backend (main)
$ npm test

> taskboard-backend@1.0.0 test
> jest --detectOpenHandles --forceExit

 PASS  tests/tasks.test.js
 PASS  tests/health.test.js

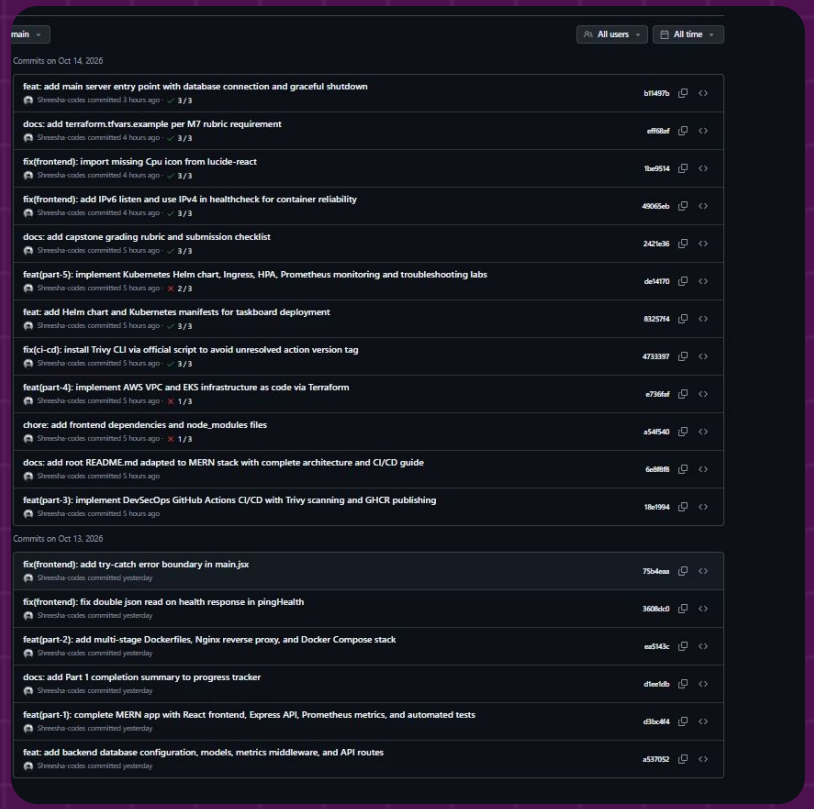
Test Suites: 2 passed, 2 total
Tests:       10 passed, 10 total
Snapshots:   0 total
Time:        4.638 s, estimated 27 s
Ran all test suites.
```

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Semantic commits make delivery auditable

- Public GitHub repository with conventional feat, fix, and docs history
- 10+ granular commits track iterative delivery across milestones
- Strict .gitignore policy prevents secret and state-file leakage

PLACEHOLDER · GitHub commit-history screenshot



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Docker turns the stack into one repeatable command

01

Build

Multi-stage frontend build keeps the delivery image focused.

02

Harden

Node.js backend runs as an unprivileged non-root user.

03

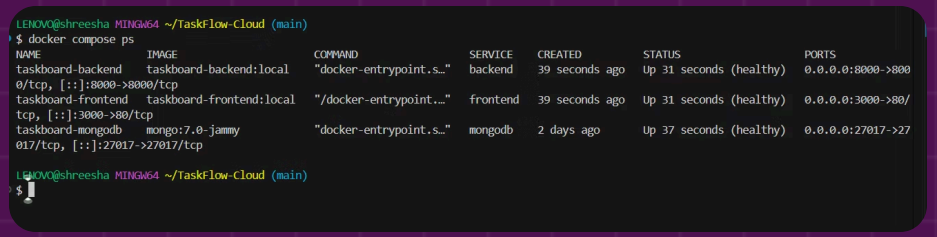
Coordinate

Compose connects frontend, backend, and MongoDB with health checks.

04

Start

docker compose up --build



PLACEHOLDER · healthy frontend, backend, and MongoDB terminal screenshot

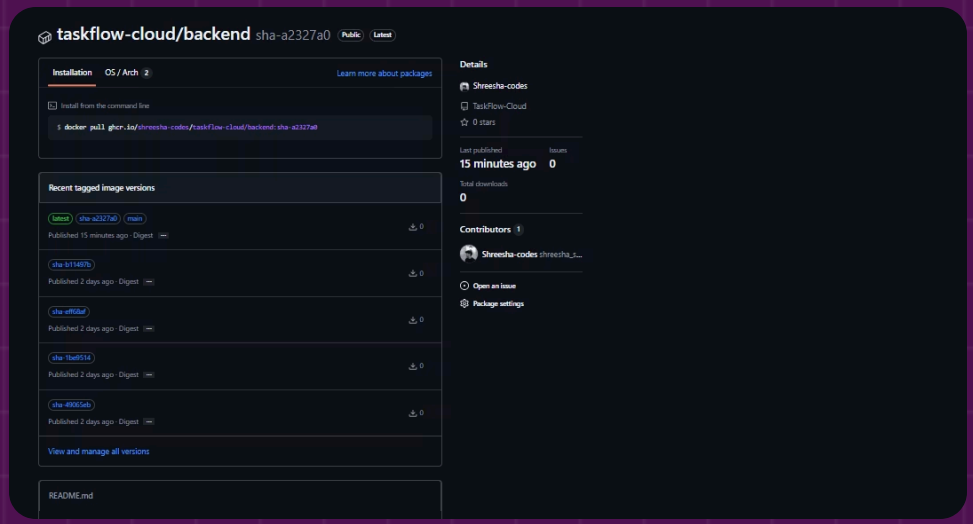
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PLACEHOLDER · GHCR package or green pipeline screenshot

05 · CI/CD

CI/CD publishes immutable releases to GHCR

- Push-to-main automation runs tests, linting, security checks, and Docker builds
- Images publish to GitHub Container Registry
- Commit-SHA tags such as sha-a2327a0 create immutable, auditable releases



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DevSecOps blocks vulnerable images before release

Security checks run inside the CI workflow.

Scan the image

Aqua Security Trivy scans the container OS layer.

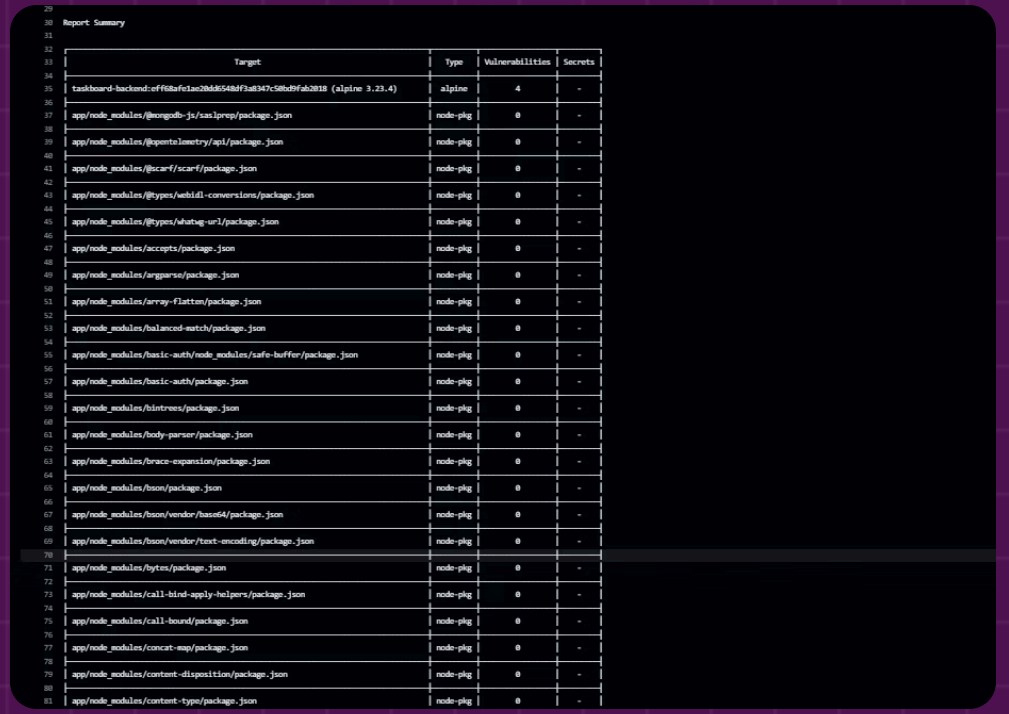
Scan dependencies

npm dependencies are included in the security check.

Block the release

Zero blocking critical vulnerabilities establishes a production-safe baseline.

PLACEHOLDER · Trivy results screenshot



Report Summary

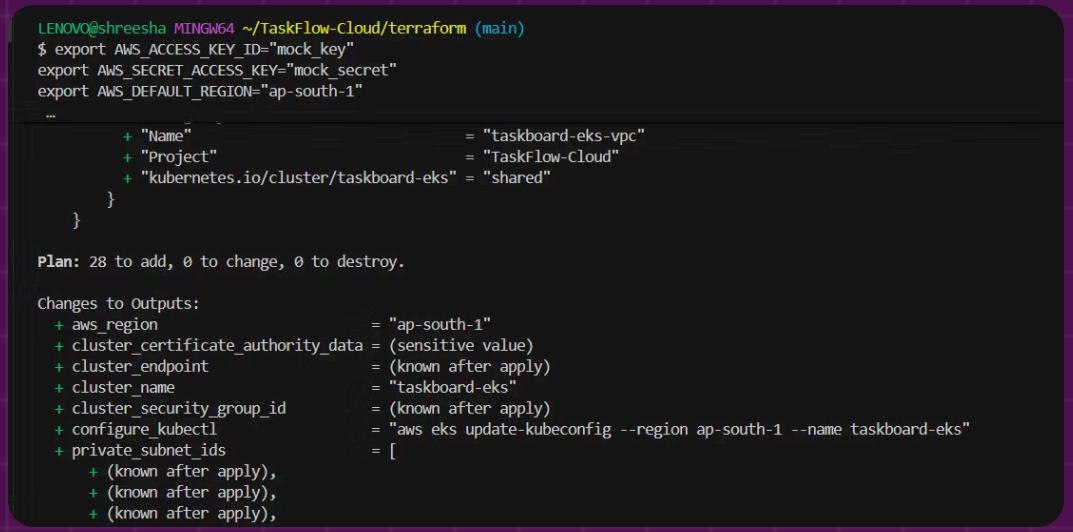
Target	Type	Vulnerabilities	Secrets
taskboard-backend:eff68afe1ac20dd6548df3a8347c58bd9fab2818 (alpine 3.23.4)	alpine	4	-
app/node_modules/@mongodb-js/saslprep/package.json	node-pkg	0	-
app/node_modules/@opentelemetry/api/package.json	node-pkg	0	-
app/node_modules/@scarf/scarf/package.json	node-pkg	0	-
app/node_modules/@types/webidl-conversions/package.json	node-pkg	0	-
app/node_modules/@types/whatwg-url/package.json	node-pkg	0	-
app/node_modules/accepts/package.json	node-pkg	0	-
app/node_modules/argparse/package.json	node-pkg	0	-
app/node_modules/array-flatten/package.json	node-pkg	0	-
app/node_modules/balanced-match/package.json	node-pkg	0	-
app/node_modules/basic-auth/node_modules/safe-buffer/package.json	node-pkg	0	-
app/node_modules/basic-auth/package.json	node-pkg	0	-
app/node_modules/bintrees/package.json	node-pkg	0	-
app/node_modules/body-parser/package.json	node-pkg	0	-
app/node_modules/brace-expansion/package.json	node-pkg	0	-
app/node_modules/bson/package.json	node-pkg	0	-
app/node_modules/bson/vendor/base64/package.json	node-pkg	0	-
app/node_modules/bson/vendor/text-encoding/package.json	node-pkg	0	-
app/node_modules/bytes/package.json	node-pkg	0	-
app/node_modules/call-bind-apply-helpers/package.json	node-pkg	0	-
app/node_modules/call-bound/package.json	node-pkg	0	-
app/node_modules/concat-map/package.json	node-pkg	0	-
app/node_modules/content-disposition/package.json	node-pkg	0	-
app/node_modules/content-type/package.json	node-pkg	0	-

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07 · INFRASTRUCTURE AS CODE

Terraform defines a highly available AWS foundation

- Modular definitions provision the VPC, subnets, and Amazon EKS
- Multi-AZ design spans **ap-south-1a**, **ap-south-1b**, and **ap-south-1c**
- **terraform plan** completes cleanly with 28 resources planned for creation



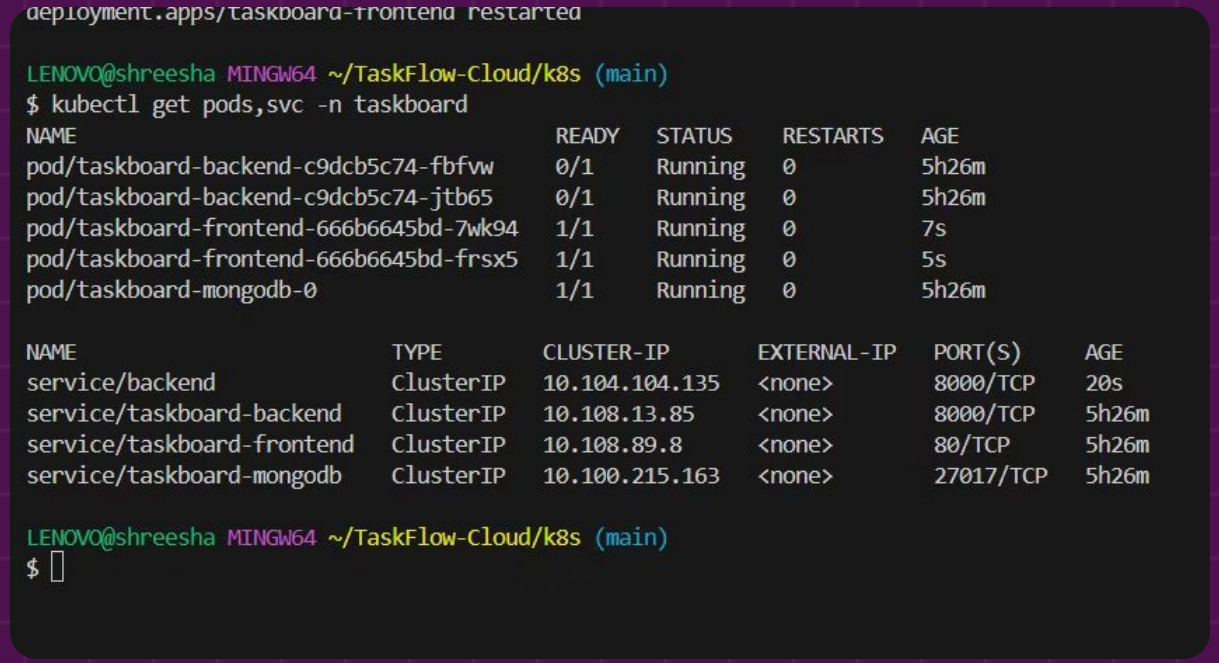
PLACEHOLDER · Terraform plan screenshot

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Kubernetes and Helm turn infrastructure into a deployable product

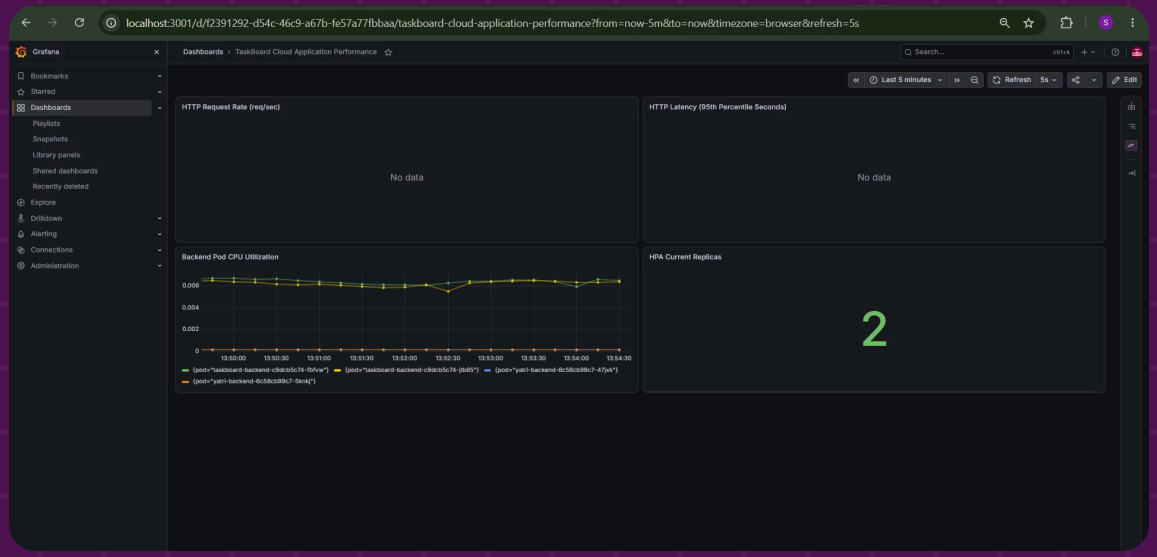
- Modular **helm/taskboard** chart manages the cluster lifecycle
- Two backend and two frontend replicas run behind ClusterIP services
- Nginx Ingress routes **http://taskboard.local** to frontend and backend services

PLACEHOLDER · Kubernetes and Helm status screenshot



Made with GAMMA

PLACEHOLDER · Grafana dashboard screenshot



09 · OBSERVABILITY

Observability closes the loop with live telemetry

Expose

/metrics through prom-client.

Scrape

Prometheus Operator uses a ServiceMonitor.

See

Grafana visualizes backend pod CPU utilization and autoscaled replicas.

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